Tutorial No : 1

Tutorial Title: Perform following String operations: a. substring b. palindrome c. compare d. copy e. reverse.
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Program:

```
#include <stdio.h>

int main()
{
	int choice;
	char str1[100], str2[100], str3[100];
	int pos, len, i, j, flag;

	printf("String Operations\n");
	printf("1. Substring\n");
	printf("2. Palindrome\n");
	printf("3. Compare\n");
	printf("4. Copy\n");
	printf("5. Reverse\n");
	printf("Choose an option (1-5): ");
	scanf("%d", &choice);

	switch(choice)
	{
	case 1:
	printf("Enter a string: ");
	scanf("%s", str1);

	printf("Enter starting position (0-based index): ");
	scanf("%d", &pos);

	printf("Enter length of substring: ");
	scanf("%d", &len);

	j = 0;

	for(i = pos; i < pos + len && str1[i] != '\0'; i++)
```

```
{
str2[j] = str1[i];
j++;
}

str2[j] = '\0';

printf("Substring: %s\n", str2);
break;

case 2:
printf("Enter a string: ");
scanf("%s", str3);

len = 0;
i = 0;
flag = 0;

while(str3[len] != '\0')
len++;

j = len - 1;

while(i < len / 2)
{
if(str3[i] != str3[j])
{
        flag = 1;
        break;
}
i++;
j--;
}

if(flag == 0)
printf("String is a palindrome.\n");
else
printf("String is not a palindrome.\n");

break;

case 3:
printf("Enter first string: ");
scanf("%s", str1);
```

```
printf("Enter second string: ");
scanf("%s", str2);

i = 0;

while(str1[i] != '\0' || str2[i] != '\0')
{
if(str1[i] != str2[i])
            break;

i++;
}

if(str1[i] == '\0' && str2[i] == '\0')
printf("Strings are matching.\n");
else
printf("Strings are not matching.\n");

break;

case 4:
printf("Enter a string: ");
scanf("%s", str1);

i = 0;

while(str1[i] != '\0')
{
str2[i] = str1[i];
i++;
}

str2[i] = '\0';

printf("Copied string: %s\n", str2);

break;

case 5:
printf("Enter a string: ");
scanf("%s", str1);

len = 0;
```

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        while(str1[len] != '\0')
        len++;

        printf("Reversed string: ");

        for(i = len - 1; i >= 0; i--)
        printf("%c", str1[i]);

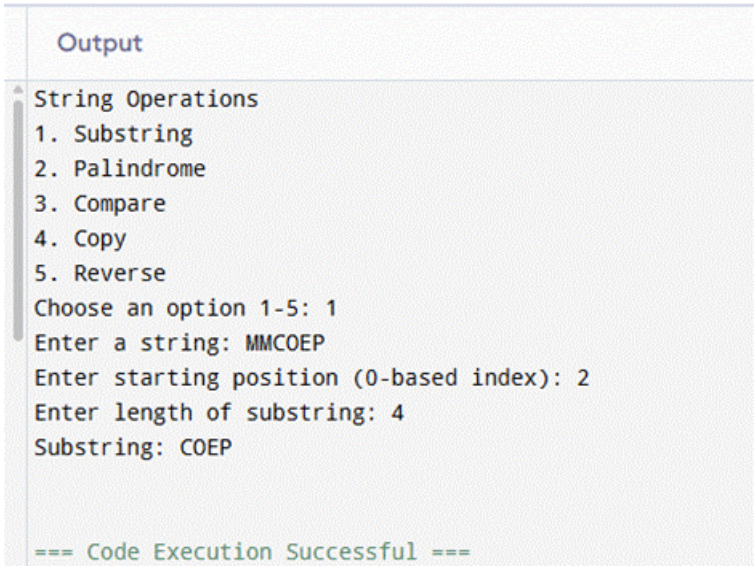
        printf("\n");

        break;

        default:
        printf("Invalid choice!\n");
        }

        return 0;
}
```

Output:



Output

```
String Operations
1. Substring
2. Palindrome
3. Compare
4. Copy
5. Reverse
Choose an option 1-5: 2
Enter a string: vijay
String is not a palindrome.


=== Code Execution Successful ===
```

Output

```
String Operations
1. Substring
2. Palindrome
3. Compare
4. Copy
5. Reverse
Choose an option 1-5: 5
Enter a string: PROGRAMIZ
Reversed string: ZIMARGORP


=== Code Execution Successful ===
```